

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration: 1 Hour

Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I T. Niranjana Kumar (192473007) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: T. Niranjana Kumar

1. Hospital Doctor Scheduling using Backtracking Search

Given Constraints

- D1 cannot work **Night**
- D2 must work **before D3** (Morning < Afternoon < Night)
- Only one doctor per shift
- D3 cannot work **Morning**
- Every doctor is assigned exactly one shift

Step 1: Assign D1

Possible shifts:

- Morning
- Afternoon
- Night

Choose **Morning**.

Doctor Shift

D1 Morning

Constraint Check:

- D1 not Night

Step 2: Assign D2

Remaining shifts:

- Afternoon
- Night

Choose **Afternoon**.

Doctor Shift

D1 Morning

D2 Afternoon

Constraint Check:

- One doctor per shift

Step 3: Assign D3

Remaining shift:

- Night

Doctor Shift

D1 Morning

D2 Afternoon

D3 Night

Constraint Check:

- D3 not Morning
- D2 before D3
- One doctor per shift
- All doctors assigned

Final Schedule

Doctor Shift

D1 Morning

D2 Afternoon

D3 Night

2. Robot Grid using BFS

Example Grid (5 × 5)

S . . X .

. X . X .

. X . . .

. . X X .

. . . . G

S = Start

G = Goal

X = Obstacle

BFS Node Expansion

Step Expanded Node Queue

1	S	(1,0),(0,1)
2	(1,0)	(0,1),(2,0)
3	(0,1)	(2,0),(0,2)
4	(2,0)	(0,2),(3,0)
5	(0,2)	(3,0),(1,2)
6	(3,0)	(1,2),(4,0),(3,1)
7	(1,2)	(4,0),(3,1),(2,2)
8	(2,2)	(2,3)
9	(2,3)	(2,4)
10	(2,4)	(3,4)
11	(3,4)	(4,4)
12	Goal Found	Stop

Optimal Path

S

↓

↓

→

→

↓

→

↓

G

or

$S \rightarrow (1,0) \rightarrow (2,0) \rightarrow (2,2) \rightarrow (2,3) \rightarrow (2,4) \rightarrow (3,4) \rightarrow G$

Total Cost

Each move costs **1**.

Total Cost = 8

Manhattan Distance Formula

$$h(n) = |x_g - x| + |y_g - y|$$

3. Autonomous Rescue Robot

(a) Constraint Analysis

Constraints and Effects

Constraint	Effect
Move Up, Down, Left, Right	Robot moves only in four directions
Obstacles	Robot cannot cross blocked cells

Constraint	Effect
Limited sensing	Robot knows only nearby cells
Movement cost = 1	Robot prefers shorter paths
Risk zone cost = +2	Robot avoids risky areas
Online search	Robot updates its path when new information is found

(b) Solution Approach

Search Strategy

Uniform Cost Search (UCS) + Online Search Agent

Steps

1. Start at **S**.
2. Observe neighboring cells.
3. Calculate movement cost.
4. Avoid blocked cells.
5. Add risky zone cost if necessary.
6. Choose the path with the lowest total cost.
7. Update knowledge after each move.
8. Continue until **G** is reached.

(c) AI Concepts

Intelligent Agent

- Observes surroundings.
- Chooses the best action.
- Learns new information while moving.

Rationality

- Selects the path with the **minimum total cost**.
- Avoids obstacles and risky zones.

Environment Type

- **Partially Observable** (limited sensing)
- **Dynamic** (new obstacles may appear)
- **Sequential** (current action affects future actions)
- **Unknown** (environment discovered while moving)

(d) Justification

Uniform Cost Search (UCS) is suitable because:

- It always finds the **minimum-cost path**.
- It considers both movement and risk costs.
- It works well when path costs are different.
- Combined with an **Online Search Agent**, the robot can adapt to newly discovered obstacles and update its route dynamically.